

Nightingale Platform API Documentation

Version 1.2 - REST API Reference

Table of Contents

1. [Getting Started](#)
 2. [Authentication](#)
 3. [Core Concepts](#)
 4. [API Endpoints](#)
 5. [Webhooks](#)
 6. [Rate Limits](#)
 7. [Error Handling](#)
 8. [Code Examples](#)
-

Getting Started

Base URL

Production: <https://api.nightingale.health/v1>
Sandbox: <https://sandbox-api.nightingale.health/v1>

API Version

Current version: **v1.2** (Released: March 2024)

The Nightingale API uses semantic versioning. Breaking changes will result in a new major version (v2, v3, etc.). We maintain backward compatibility for at least 12 months after introducing breaking changes.

Content Type

All requests and responses use JSON format.

Content-Type: `application/json`
Accept: `application/json`

Technology Stack

API Gateway: Kong Gateway (v3.2) running on AWS API Gateway **Backend Services:** Go (Golang 1.21) microservices **Request Processing:**

- Average latency: 45ms (p50), 120ms (p99)
- Powered by: Goroutines for concurrent request handling
- Service mesh: Istio for inter-service communication

Database Layer:

- PostgreSQL 15 for relational data
- MongoDB 6.0 for flexible document storage
- Redis 7.0 for caching and session management

Message Queue: Apache Kafka for asynchronous event processing

Authentication

API Key Authentication

All API requests require authentication using an API key passed in the `Authorization` header.

http

`Authorization: Bearer YOUR_API_KEY_HERE`

Obtaining an API Key:

1. Log into the Nightingale dashboard
2. Navigate to Settings → API Keys
3. Click "Generate New Key"
4. Store securely - keys cannot be retrieved after creation

API Key Format:

`nk_live_a7f3k9m2p8q4r6s1t5u0v7w9x2y4z6`

- Prefix indicates environment: `nk_live_` (production) or `nk_test_` (sandbox)
- 32-character alphanumeric identifier

JWT Tokens (OAuth 2.0)

For user-specific actions, use JWT tokens obtained via OAuth 2.0 flow.

Token Endpoint:

http

POST /v1/auth/token

Content-Type: application/json

```
{
  "grant_type": "password",
  "username": "doctor@hospital.com",
  "password": "secure_password",
  "client_id": "your_client_id",
  "client_secret": "your_client_secret"
}
```

Response:

json

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "Bearer",
  "expires_in": 3600,
  "refresh_token": "rt_a7f3k9m2p8q4r6s1t5u0v7w9x2y4z6",
  "scope": "messages:write messages:read users:read"
}
```

Token Expiration: Access tokens expire after 1 hour. Use refresh tokens to obtain new access tokens without re-authentication.

Security Note: Tokens are signed using RSA-256. Never expose client secrets in client-side code.

Core Concepts

Resources

User: A healthcare provider (doctor, nurse, technician) with a Nightingale account **Channel:** A conversation space (1-on-1 DM, group chat, or department channel) **Message:** A communication sent within a channel **Alert:** A critical notification with escalation policy **Presence:** Real-time status of a user (online, offline, busy, away)

Object IDs

All objects have unique IDs following this format:

```
usr_a7f3k9m2p8q4r6s1 (User ID)
chn_t5u0v7w9x2y4z6b8 (Channel ID)
msg_c4d6e8f0g2h4i6j8 (Message ID)
alt_k0l2m4n6o8p0q2r4 (Alert ID)
```

IDs are prefixed to indicate object type and use base58 encoding.

Pagination

Endpoints returning lists support cursor-based pagination for optimal performance.

Request:

```
http

GET /v1/messages?channel_id=chn_abc123&limit=50&cursor=cur_xyz789
```

Response:

```
json

{
  "data": [ /* array of messages */ ],
  "pagination": {
    "next_cursor": "cur_def456",
    "has_more": true,
    "total_count": 1247
  }
}
```

Parameters:

- `limit`: Number of results per page (default: 50, max: 100)
- `cursor`: Pagination cursor from previous response

API Endpoints

Messages

Send a Message

http

POST /v1/messages

Send a new message to a channel.

Request Body:

json

```
{
  "channel_id": "chn_a7f3k9m2p8q4r6s1",
  "content": "Patient in Room 302 is showing signs of tachycardia. BP 160/95.",
  "priority": "high",
  "metadata": {
    "patient_id": "pt_12345",
    "room_number": "302"
  },
  "attachments": [
    {
      "type": "image",
      "url": "https://cdn.nightingale.health/ecg_12345.jpg",
      "filename": "ecg_reading.jpg"
    }
  ]
}
```

Parameters:

- `channel_id` (required): Target channel ID
- `content` (required): Message text (max 4000 characters)
- `priority` (optional): "normal" (default), "high", "critical"
- `metadata` (optional): Custom key-value pairs for integration
- `attachments` (optional): Array of file attachments

Response (201 Created):

json

```
{
  "id": "msg_c4d6e8f0g2h4i6j8",
  "channel_id": "chn_a7f3k9m2p8q4r6s1",
  "author_id": "usr_x9y0z1a2b3c4d5e6",
  "content": "Patient in Room 302 is showing signs of tachycardia. BP 160/95.",
  "priority": "high",
  "created_at": "2024-03-15T14:32:18Z",
  "edited_at": null,
  "metadata": {
    "patient_id": "pt_12345",
    "room_number": "302"
  },
  "attachments": [
    {
      "id": "att_f7g8h9i0j1k2",
      "type": "image",
      "url": "https://cdn.nightingale.health/ecg_12345.jpg",
      "filename": "ecg_reading.jpg",
      "size_bytes": 245678
    }
  ],
  "reactions": [],
  "thread_count": 0
}
```

Code Example (Python):

python

```
import requests

url = "https://api.nightingale.health/v1/messages"
headers = {
    "Authorization": "Bearer nk_live_your_api_key",
    "Content-Type": "application/json"
}
payload = {
    "channel_id": "chn_a7f3k9m2p8q4r6s1",
    "content": "Patient vitals updated",
    "priority": "high"
}

response = requests.post(url, json=payload, headers=headers)
print(response.json())
```

Code Example (JavaScript/Node.js):

javascript

```
const axios = require('axios');

async function sendMessage() {
  try {
    const response = await axios.post(
      'https://api.nightingale.health/v1/messages',
      {
        channel_id: 'chn_a7f3k9m2p8q4r6s1',
        content: 'Patient vitals updated',
        priority: 'high'
      },
      {
        headers: {
          'Authorization': 'Bearer nk_live_your_api_key',
          'Content-Type': 'application/json'
        }
      }
    );
    console.log(response.data);
  } catch (error) {
    console.error('Error:', error.response.data);
  }
}

sendMessage();
```

Code Example (cURL):

bash

```
curl -X POST https://api.nightingale.health/v1/messages \
-H "Authorization: Bearer nk_live_your_api_key" \
-H "Content-Type: application/json" \
-d '{
  "channel_id": "chn_a7f3k9m2p8q4r6s1",
  "content": "Patient vitals updated",
  "priority": "high"
}'
```

Retrieve Messages

http

GET /v1/messages?channel_id={channel_id}

Retrieve messages from a specific channel.

Query Parameters:

- `channel_id` (required): Channel to retrieve messages from
- `limit` (optional): Number of messages (default: 50, max: 100)
- `cursor` (optional): Pagination cursor
- `since` (optional): ISO 8601 timestamp - only return messages after this time
- `priority` (optional): Filter by priority ("normal", "high", "critical")

Response (200 OK):

json

```
{
  "data": [
    {
      "id": "msg_c4d6e8f0g2h4i6j8",
      "channel_id": "chn_a7f3k9m2p8q4r6s1",
      "author_id": "usr_x9y0z1a2b3c4d5e6",
      "content": "Patient in Room 302 is showing signs of tachycardia.",
      "priority": "high",
      "created_at": "2024-03-15T14:32:18Z"
    }
  ],
  "pagination": {
    "next_cursor": "cur_def456",
    "has_more": true
  }
}
```

Update a Message

http

PATCH /v1/messages/{message_id}

Edit an existing message. Only the original author can edit messages.

Request Body:

```
json

{
  "content": "Updated: Patient in Room 302 is stable. BP now 135/85."
}
```

Response (200 OK):

```
json

{
  "id": "msg_c4d6e8f0g2h4i6j8",
  "content": "Updated: Patient in Room 302 is stable. BP now 135/85.",
  "edited_at": "2024-03-15T14:45:22Z"
}
```

Delete a Message

```
http

DELETE /v1/messages/{message_id}
```

Soft-delete a message. Message content is replaced with "[Message deleted]" but metadata is retained for audit logs.

Response (204 No Content)

Channels

Create a Channel

```
http

POST /v1/channels
```

Create a new communication channel (group chat or 1-on-1).

Request Body:

json

```
{
  "name": "Cardiology On-Call",
  "type": "group",
  "members": [
    "usr_a1b2c3d4e5f6",
    "usr_g7h8i9j0k1l2",
    "usr_m3n4o5p6q7r8"
  ],
  "description": "On-call communication for cardiology team",
  "metadata": {
    "department": "cardiology",
    "shift": "night"
  }
}
```

Parameters:

- `name` (required): Channel name (max 80 characters)
- `type` (required): "direct" (1-on-1) or "group"
- `members` (required): Array of user IDs to add
- `description` (optional): Channel description
- `metadata` (optional): Custom key-value pairs

Response (201 Created):

json

```
{
  "id": "chn_s4t5u6v7w8x9y0z1",
  "name": "Cardiology On-Call",
  "type": "group",
  "created_by": "usr_a1b2c3d4e5f6",
  "created_at": "2024-03-15T14:50:00Z",
  "member_count": 3,
  "description": "On-call communication for cardiology team"
}
```

List Channels

http

GET /v1/channels

List all channels the authenticated user is a member of.

Query Parameters:

- `type` (optional): Filter by "direct" or "group"
- `limit` (optional): Results per page (default: 50)
- `cursor` (optional): Pagination cursor

Response (200 OK):

```
json
{
  "data": [
    {
      "id": "chn_s4t5u6v7w8x9y0z1",
      "name": "Cardiology On-Call",
      "type": "group",
      "member_count": 3,
      "last_message_at": "2024-03-15T15:20:30Z",
      "unread_count": 5
    }
  ],
  "pagination": {
    "next_cursor": null,
    "has_more": false
  }
}
```

Users

Get User Information

http

GET /v1/users/{user_id}

Retrieve information about a specific user.

Response (200 OK):

json

```
{
  "id": "usr_a1b2c3d4e5f6",
  "email": "dr.chen@hospital.com",
  "display_name": "Dr. Sarah Chen",
  "title": "Attending Physician",
  "department": "Emergency Medicine",
  "phone": "+1-415-555-0123",
  "status": "online",
  "avatar_url": "https://cdn.nightingale.health/avatars/usr_a1b2c3d4e5f6.jpg",
  "timezone": "America/Los_Angeles",
  "created_at": "2023-06-15T09:00:00Z",
  "metadata": {
    "employee_id": "EMP-12345",
    "specialty": "emergency_medicine"
  }
}
```

Get User Presence

http

GET /v1/users/{user_id}/presence

Get real-time presence status of a user.

Response (200 OK):

json

```
{
  "user_id": "usr_a1b2c3d4e5f6",
  "status": "online",
  "custom_status": "On call - ED",
  "last_seen": "2024-03-15T15:45:12Z",
  "is_on_call": true,
  "device": "mobile"
}
```

Status Values:

- `online`: Actively using the application
 - `away`: Inactive for 10+ minutes
 - `busy`: Do not disturb mode enabled
 - `offline`: Not connected
-

Alerts

Create an Alert

http

POST /v1/alerts

Create a critical alert with escalation policy.

Request Body:

json

```
{
  "title": "Critical Lab Result - Patient Johnson",
  "message": "Troponin levels elevated to 2.5 ng/mL. Immediate cardiology consult required.",
  "severity": "critical",
  "recipient_id": "usr_g7h8i9j0k1l12",
  "escalation_policy": {
    "timeout_seconds": 300,
    "escalate_to": "usr_m3"
```